

Lab Practical Sheet: Simulating RESTful Web Services Using a Database Table

Sample Database Table: students

id	name	age	grade
1	Alice	20	A
2	Bob	22	B
3	Charlie	21	C

Instructions: You are given a students table representing a backend database. Imagine you are building a REST API with base URI:

`https://api.school.com/students`

For each task:

1. Identify and write the appropriate **HTTP method**.
 2. Construct the correct **URI**.
 3. Write the **request body** (if needed).
 4. Write the **expected server response body**.
 5. Provide the correct **HTTP status code**.
-

Tasks:

1. Retrieve all student records.
2. Get the record of the student with ID 2.
3. Add a new student named "Dana", aged 23, grade "A".
4. Update the grade of student ID 3 to "B".
5. Delete the student with ID 1.
6. Replace student ID 2's full record with: name "Ben", age 23, grade "A".
7. Try to retrieve a student with ID 99 (nonexistent).
8. Submit a bad request (e.g., POST without required fields).

1. Method: GET

URI: /students

Request Body: None

Response Body:

```
[  
  { "id": 1, "name": "Alice", "age": 20, "grade": "A" },  
  { "id": 2, "name": "Bob", "age": 22, "grade": "B" },  
  { "id": 3, "name": "Charlie", "age": 21, "grade": "C" }  
]
```